

Stochastic Modeling and Thermodynamics of RNA Folding

Heat Renaturation in Modified Biosensor Geometry

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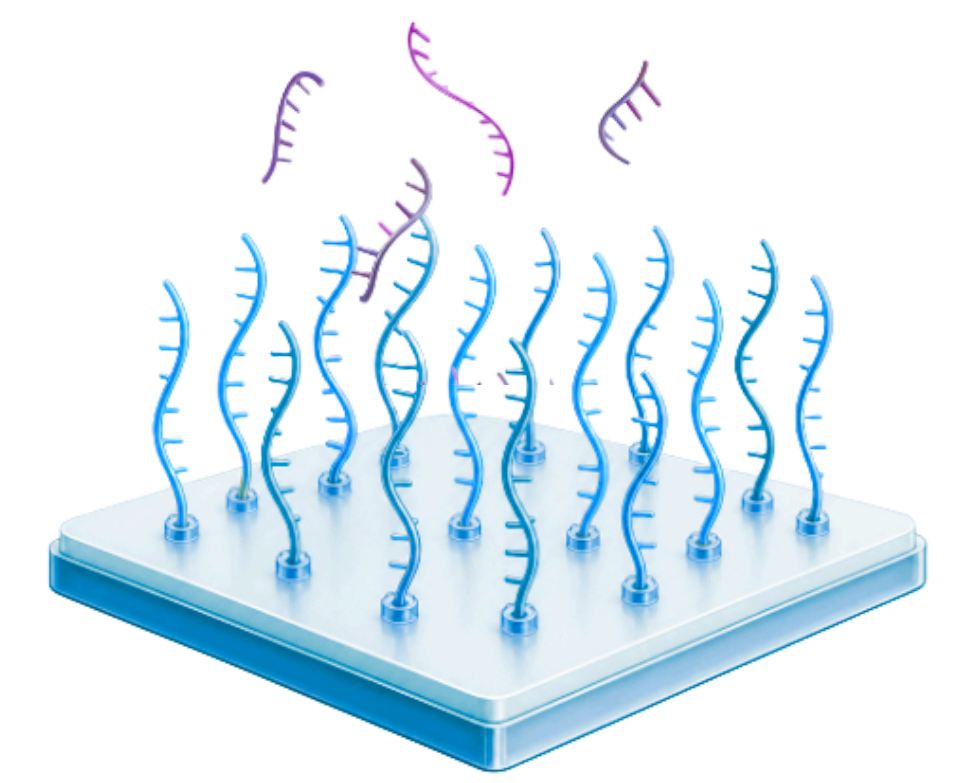
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1 Abstract

Surface-based RNA biosensors detect specific transcripts through hybridization between immobilized probes and target strands. Their behavior is governed by two physical effects: the folding of RNA chains into secondary structures, and the geometric organization of the probe brushes that are typically modeled in isolation. This project examines how these two effects combine and whether they can be treated as statistically independent. The broader goal is to clarify when decoupled descriptions of biosensor behavior are sufficient and when a unified model of folding and geometry is needed.

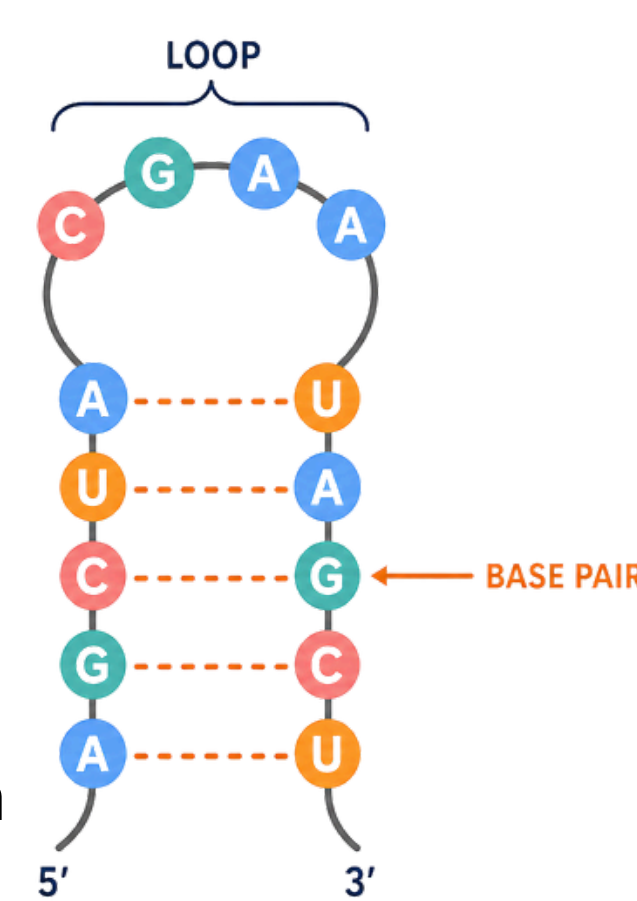
2 Introduction

A **biochip** is a sensor surface densely packed with single-stranded nucleic-acid probes. Targets in solution hybridize with their complementary probes, producing a measurable signal which is the basis of gene-expression profiling and disease diagnostics. A common issue is the probes folding onto themselves and lowering the sensor efficiency. To counter this, the chips usually create small electric fields, which penalise self-folding. The folding in these systems has been modeled thermodynamically, taking into account the quenched disorder of the sequence. We want to improve the accuracy of the predictions of these models by directly calculating the geometry of the RNA.



3 Background

- A single-stranded chain of nucleotides drawn from a four-letter alphabet: **A, U, G, C**.
- Complementary letters bind by hydrogen bonds — A–U and G–C.
- When the chain folds back on itself, complementary regions zip together: paired regions form stems, unpaired regions form loops and hairpins.
- The full pattern of which bases pair with which is the chain's **secondary structure**.



Free energy of folding: $-\beta F = \log \langle e^{-\beta E} \rangle$

The averaging is over all secondary structures. Computed via a sequence-based partition-function algorithm [1].

Helicity θ Equilibrium fraction of paired bases. Plotted against temperature T , it describes the chain's heat-driven folding/unfolding behavior.

Thermodynamic model [2]

$$E(M, \{h\}) = \sum_{ij} \epsilon_{ij} m_{ij} - \alpha D$$

E - Energy of a single RNA configuration
 M - Connection matrix (secondary structure)
Energy cost of a bond - $\epsilon_{ij} = \epsilon h_i h_j + \epsilon_0$
 $\{h\}$ - Sequence of nucleotide types
 α - The electrostatic coupling parameter
 D - Average height of the chain

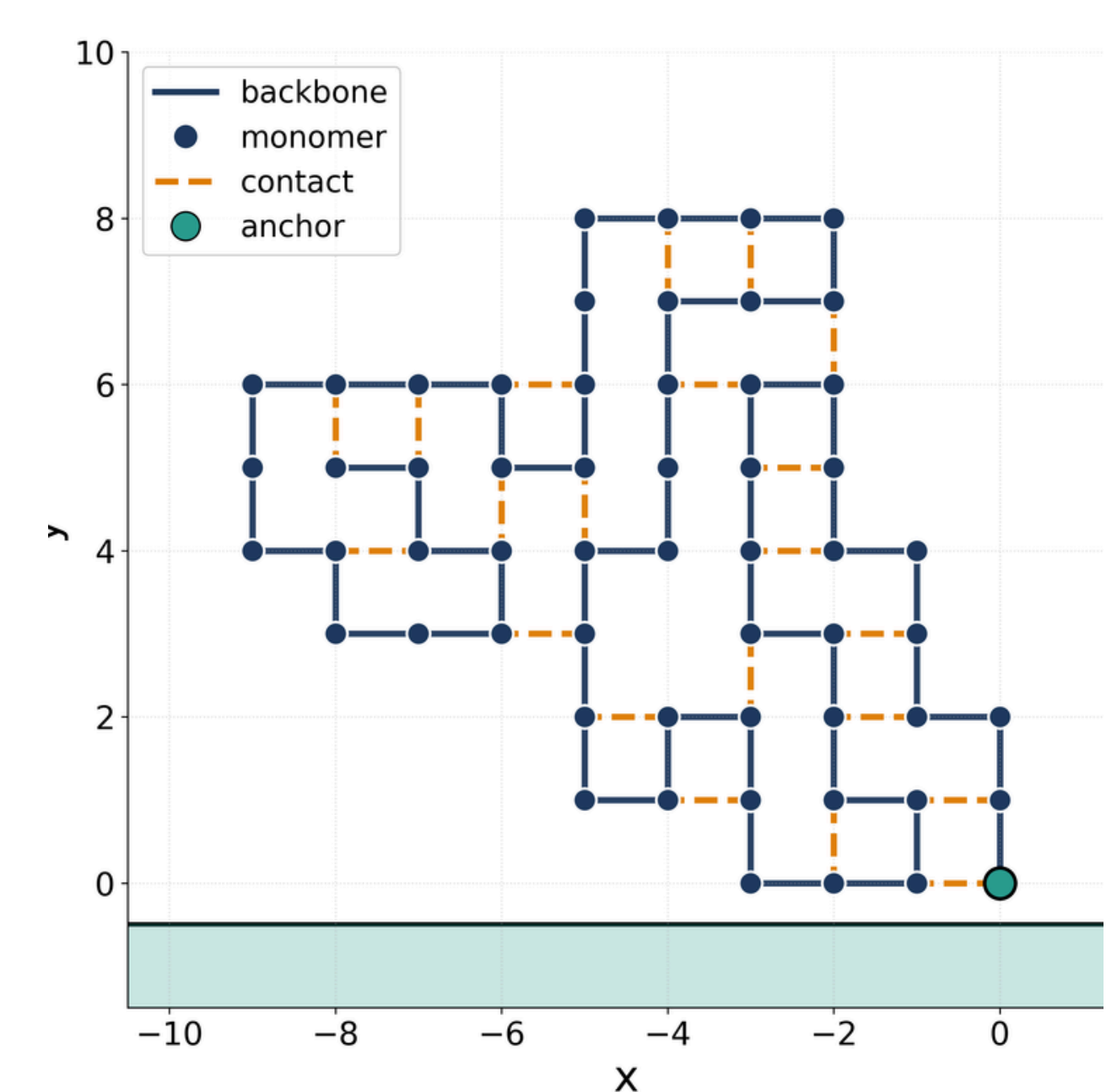
A prior work substituted ladder distance (l) for average height (D).

$$D \rightarrow 1 - l$$

Our approach was to use a random walk to model the geometry of the chain.

Self-avoiding random walks (SARW)

Lattice model of an RNA where bonds are made in monomer pairs with fixed distance



4 Methods

Sequence-based folding free energy is computed using a partition-function method [1]. Our approach is to make the electric field parameter a responsive to chain geometry.

Step 1: Simulate every polymer geometry using a self-avoiding random walk and gather a sample of fractions r .

$$r = \frac{D}{1 - l}$$

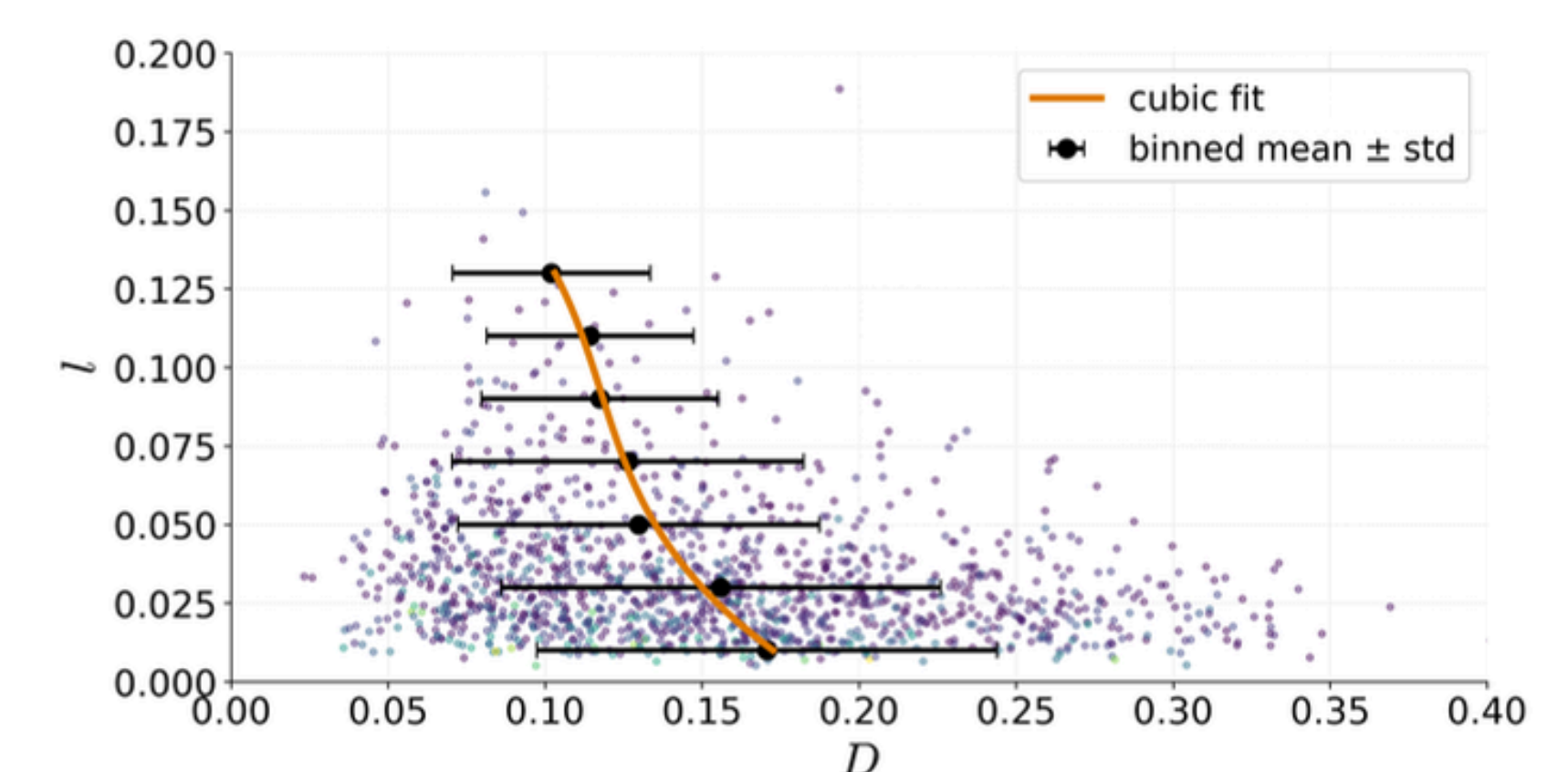
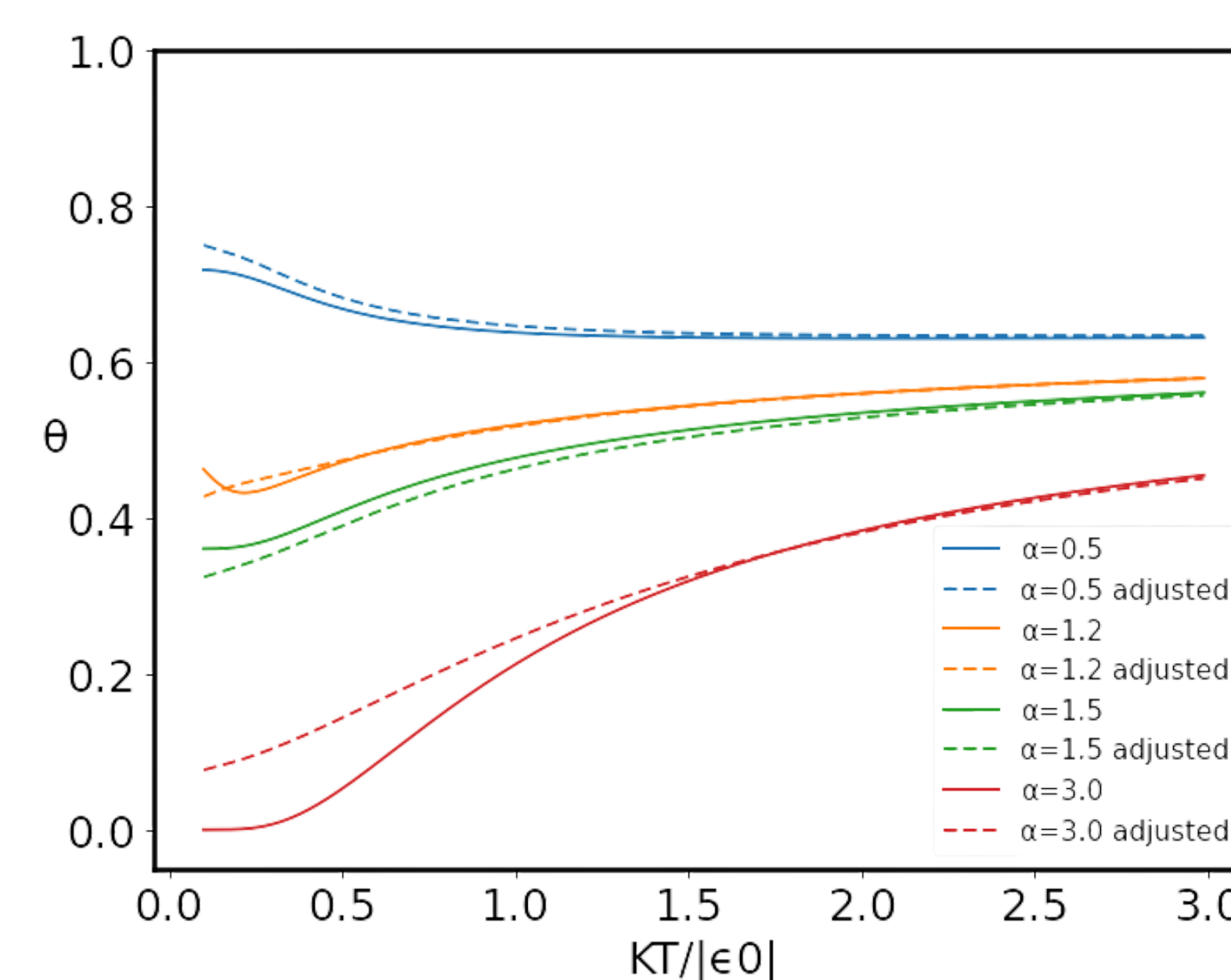
Step 2: Add the geometric dependence to the electrostatic coupling parameter to recover the prior approximation for average height.

$$\alpha \rightarrow r\alpha$$

Step 3: We find the average free energy over the sample of fractions r . Here, for simplicity, we assume that the distribution over the geometry and the distribution over the states are independent.

5 Results

SARW simulations show that the substitution $D \rightarrow 1 - l$ of correctly captured the negative trend, however, the actual data shows a non-trivial non-linear dependence of these two indicators (l , D)



The free energy averaging over our simulation sample matches the prior results for heat renaturation; however, there is a divergence in the cold temperature regimes. The overall $\theta(T)$ shape remains consistent.

6 Discussion

- The two methods agree at higher temperatures, but disagree in the colder regions.
- Although there are differences in numbers, both the prior work and the geometrically recovered model show the same pattern of folding.
- For biosensors, this folding behavior can result in unwanted efficiency loss. Both approaches show that this is more drastic for stronger electrostatic fields.
- The residual mismatch cannot, on its own, tell us whether the correction moves the result toward or away from the true coupled answer — because the test relies on the very assumption it tries to evaluate.

7 Conclusions & Future Work

Conclusions

- Built a pipeline coupling SAW geometry with sequence-based folding via $D / (1 - l)$.
- Bare and geometric helicities agree across the (α, T) grid.
- Validates protocol consistency, not the underlying independence assumption
- The simple isolated models show disagreement, which indicates that a more complex model is required to uncover the behavior.

Future Work

- Joint sampling of geometry and folding (single Monte Carlo ensemble).
- Extend SAWs to 3D and to longer chains.

[1] Bundschuh, R., & Hwa, T. (2002). Statistical mechanics of secondary structures formed by random RNA sequences. *Physical Review E*, 65(3), 031903.

[2] Hayrapetyan, G. N., Iannelli, F., Lekscha, J., Morozov, V. F., Netz, R. R., & Mamasakhlisov, Y. S. (2014). Reentrant melting of RNA with quenched sequence randomness.